Wind Tunnel Logger

Student user manual

COMP0219 Sensors and Signal Conditioning - 2026/2027 wind tunnel setup

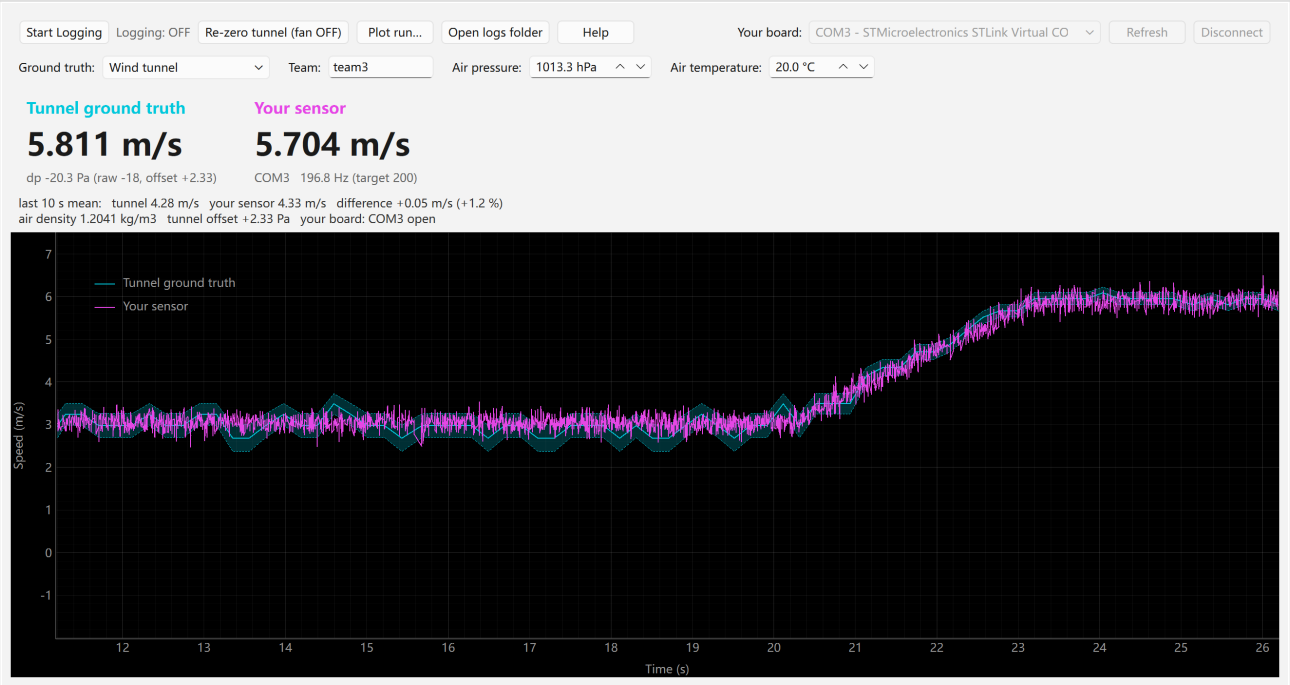


Figure 1: Wind Tunnel Logger during a tunnel run (example data).

Version 1.0.0 - Author: Shengyang Zhuang - October 2026

Live plot and logging of your airspeed sensor against the wind tunnel's own airspeed reading.

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1. What Wind Tunnel Logger does

Wind Tunnel Logger shows your sensor's airspeed next to the wind tunnel's own airspeed reading, live, and saves both to CSV files. The tunnel's reading is the **ground truth**. The lab has two tunnels, and each has its own ground truth, which you plug into your laptop by USB:



White tunnel *(TecQuipment): its pressure sensor, through its VDAS box. Pick* ***Ground truth: Wind tunnel****.*



Wood tunnel*: a GIGA pitot tube (Arduino GIGA R1 on top, its probe in the tunnel). Pick* ***Ground truth: GIGA pitot tube****.*

Figure 2: The two tunnels and their ground truth.

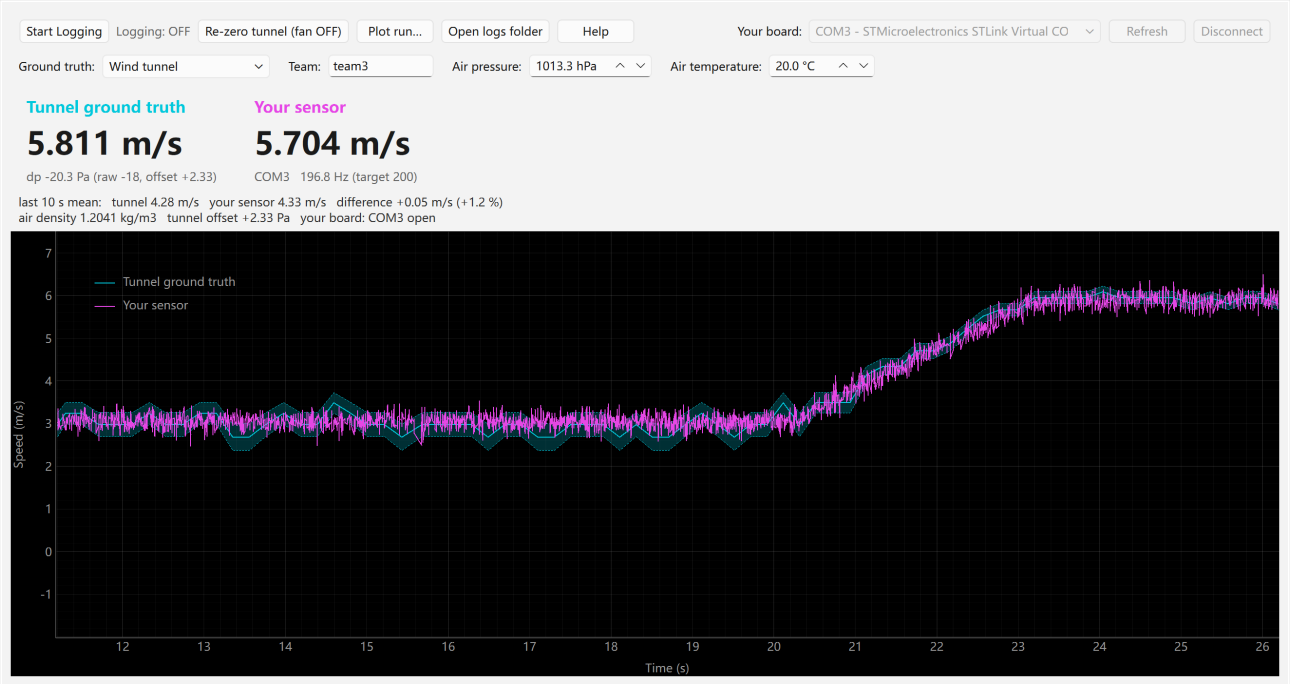


Figure 3: The main window during a run (example data).

The window, from top to bottom:

- **Toolbar**: Start Logging, Re-zero tunnel, Plot run, Open logs folder, Help, and on the right **Your board** (your sensor's port) with Refresh and Connect.
- **Settings row**: Ground truth (Wind tunnel, Simulated, or a GIGA pitot tube while one is plugged in, see section 6), your Team name, today's Air pressure and Air temperature.
- **Readouts**: the tunnel's speed (cyan) and your sensor's speed (magenta). Under your reading: your port and the **rate** your board achieves (it must reach 200 Hz).
- **Status lines**: what the program is doing and anything that needs your attention, in red.

- **Live plot**: the last 15 s. The shaded cyan band is the range the true speed can be in, given the tunnel sensor's resolution of 1 Pa; it is wide at low speed (see section 7).

What your board must do

- Appear as a USB serial port on the laptop (a NUCLEO does: *STMicroelectronics STLink Virtual COM Port*).
- Print **one speed in m/s per line**, as a plain number, for example `5.432`.
- Stream at **200 Hz**, at **115200 baud**.

```
/* in your 200 Hz loop, once per estimate (v in m/s) */
printf("%.3f\r\n", v);
```

2. What you need

Item	Why
A laptop	Windows 10 or 11 (tested). macOS and Linux should also work with the pip version; see *Using a Mac or Linux* in section 6.
Internet, once	To download Wind Tunnel Logger.
Your board and its USB cable	A data cable, not a charge-only cable.
White tunnel: its VDAS box's USB cable	USB-B (printer type) at the box end; it stays at the tunnel.
Wood tunnel: the GIGA's USB cable	USB-C at the GIGA end; it stays at the tunnel.

> **TIP** You **don't need TecQuipment's VDAS software**: Wind Tunnel Logger talks to the white tunnel's VDAS box, and to the wood tunnel's GIGA, directly. If the VDAS software is open on your laptop, close it.

3. Install Wind Tunnel Logger

Both versions are on the course GitHub **Releases** page, under *Assets* of the latest release:
`https://github.com/Dkaka/COMP0219_Tutorials/releases`

Option A - Windows program (recommended, no Python needed)

1. Download **WindTunnelLogger.exe** and move it somewhere you'll find it again, for example your Desktop or *Documents*.
2. Double-click it. The window takes a few seconds to appear: the program unpacks itself each time it starts.
3. If Windows shows "*Windows protected your PC*", click **More info**, then **Run anyway**. This appears because the program isn't code-signed; it is safe to run.

> **TIP** If your antivirus moves the file to quarantine, restore it or use Option B. Programs packaged with Python are sometimes flagged by mistake.

Option B - pip install (Windows, macOS or Linux, Python 3.9 or newer)

In a terminal (PowerShell on Windows, Terminal on a Mac), run:

```
python -m pip install https://github.com/Dkaka/COMP0219_Tutorials/releases/download/
    wind-tunnel-logger-v1.0.0/wind_tunnel_logger-1.0.0-py3-none-any.whl
```

Type it as **one line** (it is only wrapped here to fit the page). Or download `wind_tunnel_logger-1.0.0-py3-none-any.whl` from the Releases page and run `python -m pip install wind_tunnel_logger-1.0.0-py3-none-any.whl` in the folder you saved it to. On macOS or Linux, type `python3` instead of `python`. Then start the program with:

```
wind-tunnel-logger
```

If that command isn't found, use `python -m wind_tunnel_logger`. To update later, install the newer `.whl` file the same way.

4. Prepare your board

- Flash your firmware so the board prints one speed in m/s per line at 200 Hz, 115200 baud (section 1).
- Close every other program that might use the board's port: PuTTY, CoolTerm, the STM32CubeIDE serial console, an Arduino serial monitor. Only one program can open a port at a time.
- Find the port: in Windows, open **Device Manager**, then **Ports (COM & LPT)**; a NUCLEO appears as *STMicroelectronics STLink Virtual COM Port (COMx)*. On a Mac it is `/dev/tty.usbmodem...`.
- Test at home first with the simulated ground truth (section 10): you'll see straight away whether your board connects, sends numbers and reaches 200 Hz.

5. Zero the wind tunnel (every session)

WHY A pressure sensor never reads exactly 0 Pa in still air: it has a small **offset**, here a few pascals. Airspeed comes from the pressure the pitot tube sees, v = sqrt(2 Δp / ρ), and at low speed that pressure is tiny. At 2 m/s it is only about 2.4 Pa, the same size as the offset. Left in, the offset makes low speeds read far too high (table below). Zeroing measures the offset with the fan off and removes it from every reading.

True speed	Pressure the pitot sees	Reading with a 2.5 Pa offset left in	Error
2 m/s	2.4 Pa	2.86 m/s	+43 %
4.5 m/s	12.2 Pa	4.94 m/s	+10 %
10 m/s	60.2 Pa	10.21 m/s	+2 %

On the **white tunnel** there are two zeros: the tunnel's own pressure display (step A, at the start of a session) and Wind Tunnel Logger's software zero (step B, automatic every time you start it). On the **wood tunnel**, skip both: the GIGA zeroes itself when you pick it (section 6, *On the wood tunnel*), also with the fan off.

Step A - zero the tunnel's pressure display (start of a session)

Skip this step if your TA has already done it today.

1 Switch on the tunnel's instrument power and **wait 10-15 minutes**: the pressure sensor drifts while it warms up.

2 Make sure the **fan is OFF and has stopped**, nothing is blowing at the tunnel, and wait about 30 s for the air to settle.

3 On the tunnel's pressure display module (the small AFA5 module whose LCD shows the pressure), **press and hold the zero button for at least 4 seconds**. The display should then read 0 or very close to it.

IMPORTANT Never hold the zero button **while switching the power on**: that starts the module's calibration mode (the display shows - - - - or *CAL*). If it happens, switch the power off again without pressing anything else: the calibration is then left unchanged.

Step B - the software zero in Wind Tunnel Logger (automatic)

As soon as the tunnel's data arrives after the program starts (and again whenever you switch Ground truth back to Wind tunnel), it averages 5 s of tunnel readings with the fan off and removes that offset from then on. Until the data arrives, the status line shows *Waiting for the tunnel's data to zero it*. Keep the **fan OFF** until the status line stops showing *Zeroing the tunnel reading*:

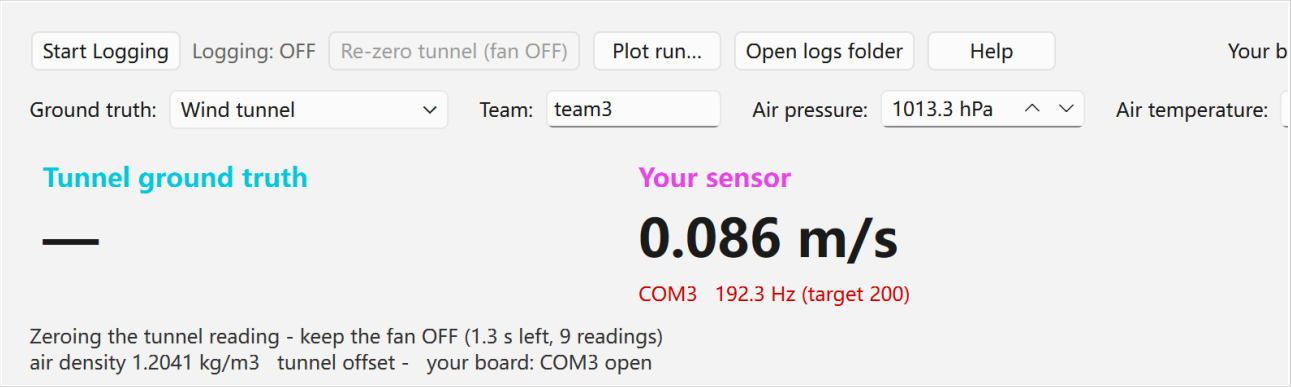


Figure 4: Zeroing in progress: keep the fan off.

When it finishes, the status line shows **tunnel offset**, normally between about -3 and +3 Pa. If the fan was running during the zero, the program can tell (the reading is far from 0), uses a typical offset of +3 Pa instead, and warns you in red:

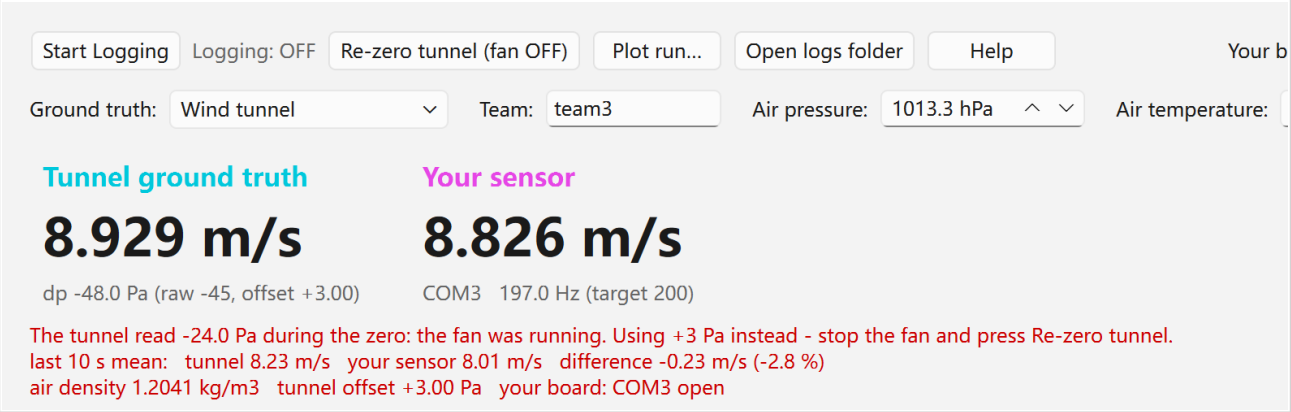


Figure 5: The zero was taken with the fan running. Stop the fan and press Re-zero tunnel.

Press Re-zero tunnel (fan OFF), with the fan stopped and the air still, whenever:

- you see that red warning;
- the tunnel's instrument power was switched off and on, or you did step A again;
- with the fan off, the tunnel reading isn't close to 0 m/s (up to about 1 m/s is normal, see section 7);
- before an important logged run, if you can stop the fan.

> **TIP** If your own sensor needs a zero too, do it at the same moment, with the fan off.

6. At the tunnel, step by step

1 **Fan OFF.** Plug the tunnel's ground truth into your laptop by USB: the **VDAS box** on the white tunnel, the **GIGA** on the wood tunnel (section 1).

2 Start Wind Tunnel Logger and pick **Ground truth**: **Wind tunnel** on the white tunnel, **GIGA pitot tube (COMx)** on the wood tunnel.

3 Within a few seconds Wind Tunnel Logger receives the tunnel's data: the status line changes from *Waiting for the tunnel's data* (or *the GIGA's data*) to *Zeroing* .... Nothing else to open: if TecQuipment's VDAS software is running, close it.

4 Wait for the zero to finish (section 5).

5 Fill in **Team** (used in your file names), today's **Air pressure** in hPa (from a lab barometer or a weather service's surface pressure) and the **Air temperature** in °C. These set the air density used to turn the tunnel's pressure into speed.

6 Plug in your board, click **Refresh**, pick its port under **Your board** and click **Connect**. Your reading starts to update; check the rate under it is close to 200 Hz and not red.

7 Start the fan at your first speed and let it settle.

8 Click **Start Logging**. Run your test: the assessed range is 2-10 m/s, including steps and ramps.

9 Click **Stop & Save**. Accept the suggested file name (team, date and time) or type your own, and click Save.

10 When you have finished, close Wind Tunnel Logger.

Until the tunnel reading arrives, the status line says what is missing, for example when the VDAS box isn't plugged in or the tunnel's instrument power is off:

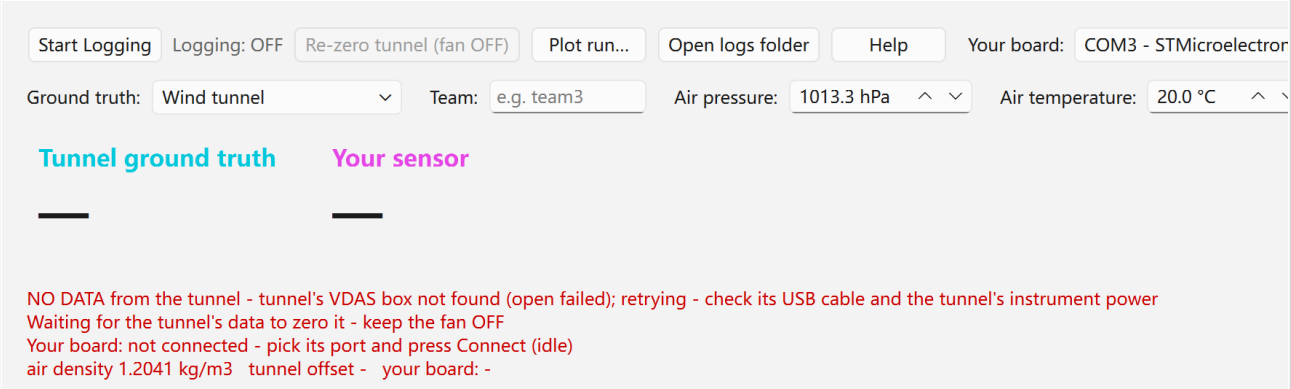


Figure 6: No tunnel data yet: the status line says why.

On the wood tunnel (GIGA pitot tube)

The wood tunnel's ground truth is a WindSpeedGiga pitot tube: an Arduino GIGA R1 whose pitot probe sits in the tunnel. While it is plugged in, **Ground truth** also offers **GIGA pitot tube (COMx)**, and its port is no longer listed under **Your board**. Keep the fan OFF when you pick it:

- The program sends the GIGA its calibration (**LEQ** 102.177 m and **C** 1, from `config.yaml`) with today's air pressure and temperature. The GIGA forgets LEQ whenever it loses power, and then reads 50-60 % low: so the program uses none of its readings until the GIGA confirms these values, and sends them again whenever it restarts.
- The GIGA zeroes itself on the board in about 2 s (**Re-zero GIGA** repeats it). The fan must be off, or the GIGA's two tubes pushed into each other.
- Its calibration is within about ±3 % at 8-14 m/s and was only checked from 4 to 14 m/s. Above its sensor's range (125 Pa, about 14 m/s) its readings aren't used.

- Saved runs end in `_giga`.

Using a Mac or Linux

Install with pip (section 3, option B). Wind Tunnel Logger reads the VDAS box on macOS and Linux too, but it has only been tested on Windows: try it at home in simulated mode (section 10), and at the tunnel. On Linux, reading USB devices needs a permission rule (section 11). If it doesn't work, plug your board into a teammate's Windows laptop, run Wind Tunnel Logger there and copy the saved files afterwards.

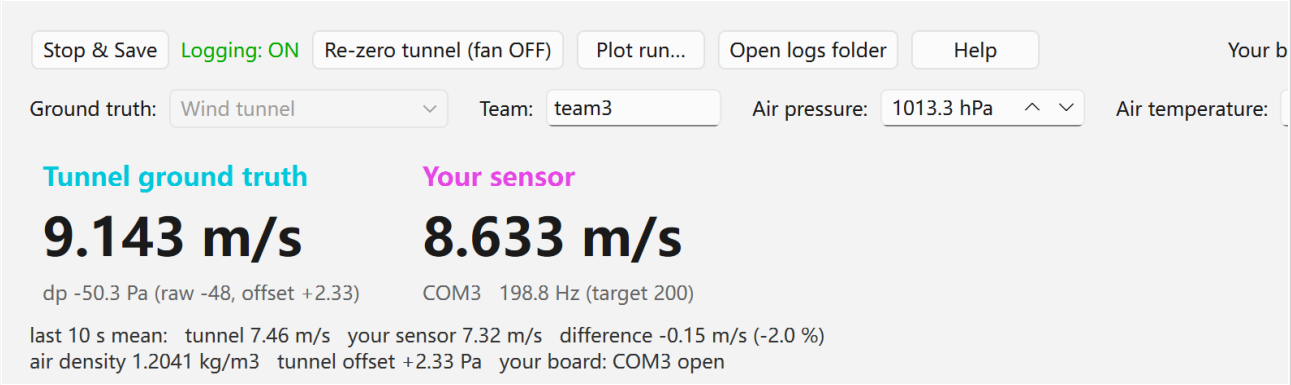


Figure 7: Logging: the button now reads Stop & Save.

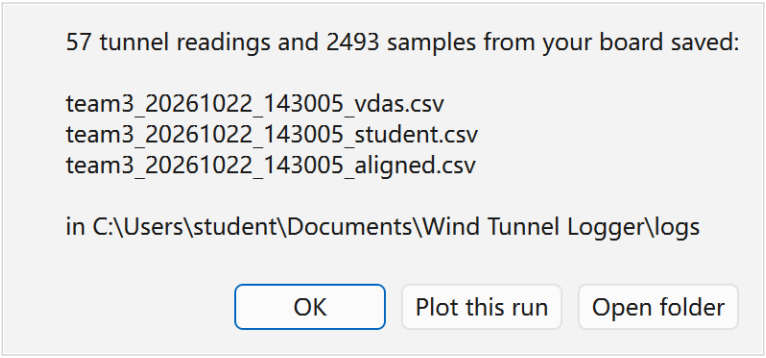


Figure 8: After Stop & Save: the three files of the run. Plot this run opens the plots of section 8.

7. Reading the screen

What you see	Meaning
Tunnel ground truth m/s	The tunnel's airspeed, from its pressure reading after the zero, using today's air density.
dp ... Pa (raw ..., offset ...)	The pressure behind that speed, the raw value from the VDAS box, and the zero offset being removed.
Your sensor m/s	The last value your board sent.
COM3 199.2 Hz (target 200)	Your port and the average rate over the last 5 s. Red if it's below 196 Hz, or if lines arrive that aren't numbers (the last bad line is shown).
last 10 s mean ... difference	Both speeds averaged over 10 s, and how far your sensor is from the tunnel.
Shaded cyan band	Where the true speed can be, given the tunnel sensor reports whole pascals (±1 Pa). It is wide at low speed because a 1 Pa step is a big change in speed there: about ±0.8 m/s at 1 m/s, ±0.2 m/s at 4.5 m/s and ±0.08 m/s at 10 m/s.
Red status lines	Something needs your attention: no data, fan running during the zero, your board not sending.

TIP With the fan off, the tunnel may show up to about 1 m/s instead of 0. That is normal: a reading of 1 Pa already corresponds to about 1.3 m/s. The same limit is why the assessment starts at 2 m/s.

8. Looking at a saved run

Plot this run (after saving) or **Plot run...** (any time; pick the run's `_aligned.csv`) opens three plots:

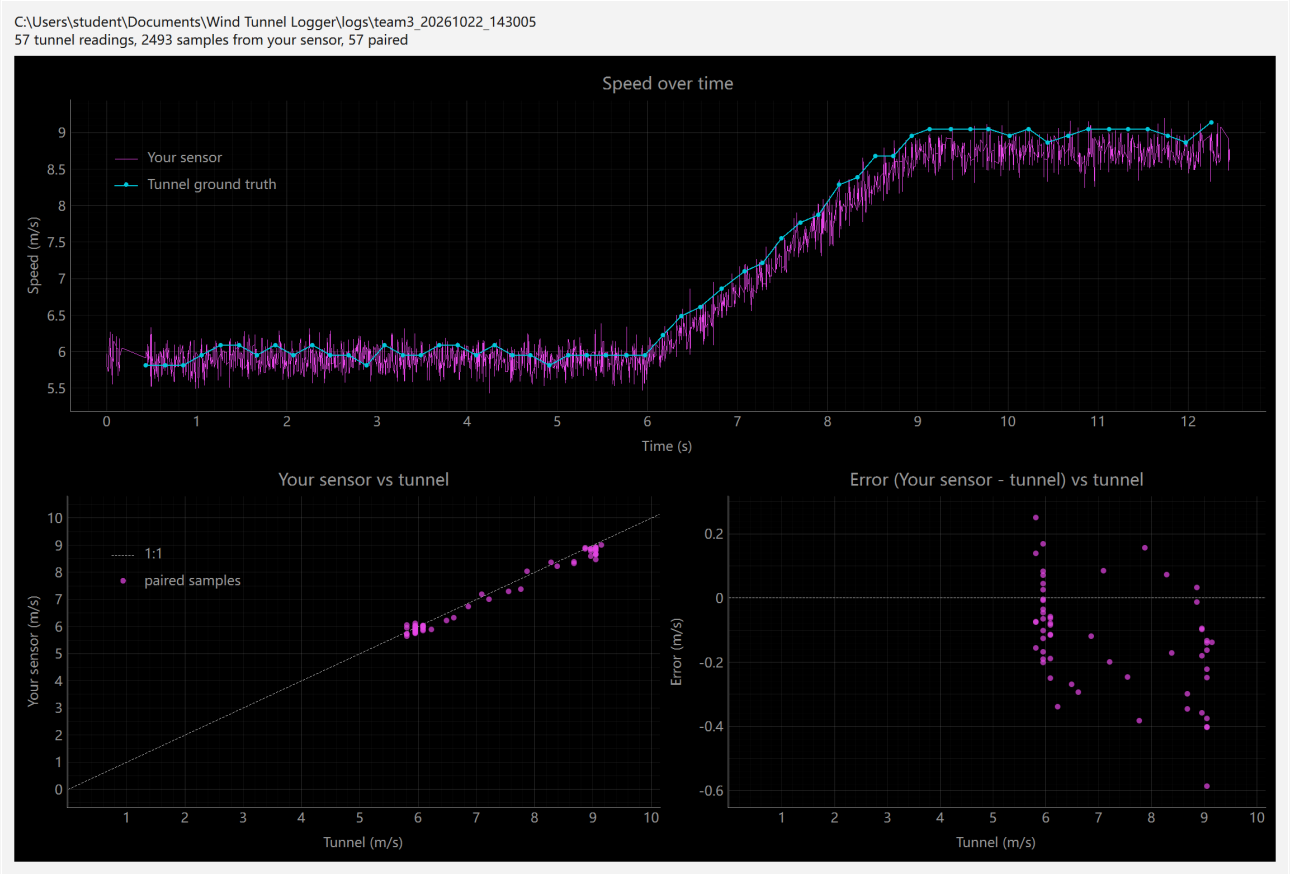


Figure 9: A saved run (example data).

- **Speed over time**: every sample of your sensor (magenta) and every tunnel reading (cyan dots, a few per second).
- **Your sensor vs tunnel**: one dot per tunnel reading, paired with your nearest sample. On the dashed 1:1 line means exactly right.
- **Error vs tunnel**: your sensor minus the tunnel, against speed. It shows at which speeds your sensor reads high or low.

9. Your saved files

Every run is saved as three CSV files in `Documents\Wind Tunnel Logger\logs` (click **Open logs folder**):

File	Contents
`team3_20261022_143005_vdas.csv`	Every tunnel reading: time, speed, raw and zero-corrected pressure.
`team3_20261022_143005_student.csv`	Every sample from your board, with the time it arrived.
`team3_20261022_143005_aligned.csv`	Each tunnel reading paired with your nearest sample (columns *ground_truth_mps* and *student_mps*). Use this one to work out your error statistics (MSE, bias, standard deviation) for the report.

Runs made with the simulated ground truth end in `_simulated`, and runs with a GIGA pitot tube as ground truth in `_giga`, so they can't be mistaken for tunnel data. Keep your files: the module team uses the same format for assessment.

10. Bench test at home (simulated ground truth)

No tunnel needed. Set **Ground truth** to **Simulated (bench test)**: the program generates a made-up tunnel speed, so you can check that your board connects, sends plain numbers, reaches 200 Hz and that logging and plotting work. The simulated speed is **not real wind**; don't compare your sensor with it.

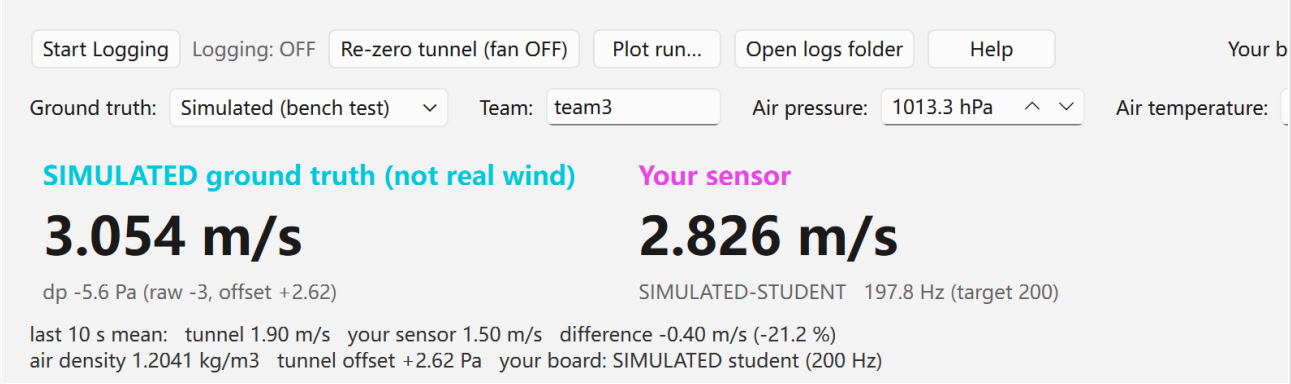


Figure 10: Simulated ground truth for bench tests.

11. Troubleshooting

Problem	What to do
Red *NO DATA from the tunnel - tunnel's VDAS box not found*	Check the VDAS box's USB cable and that the tunnel's instrument power is on. Close TecQuipment's VDAS software if it is open.
Red … *the VDAS box sends no data*	Check the tunnel's instrument power, then unplug and replug the box's USB cable. Close TecQuipment's VDAS software if it is open.
Red … *none from pressure cell 1*	The AFA5 module's instance switch isn't set to 1: ask your TA.
Linux: *VDAS box not found* although it is plugged in	Allow access to it, then unplug and replug it: `echo 'SUBSYSTEM=="hidraw", ATTRS{idVendor}=="1781", MODE="0666"' | sudo tee /etc/udev/rules.d/99-vdas.rules`, then `sudo udevadm control --reload`.
Red *the fan was running*	Stop the fan, wait for still air, press **Re-zero tunnel**.
Tunnel reads far from 0 with the fan off	Press **Re-zero tunnel**; if it stays off, do step A of section 5 (or ask your TA).
Red *The GIGA pitot tube hasn't confirmed LEQ …*	It is still starting, or isn't running the WindSpeedGiga firmware. Unplug and replug it; the end of that line shows what it reports.
Red *The GIGA rejected its zero*	Air was moving. Stop the fan (or push the GIGA's two tubes into each other) and press **Re-zero GIGA**.
cannot open COMx	Another program has the port open (PuTTY, CubeIDE console, serial monitor): close it. Or the board was unplugged: plug it in and click Refresh.
Your rate is red	Your board sends fewer than about 196 lines per second, or some lines aren't plain numbers (the bad line is shown). Check your firmware's timing and print format.
Your reading doesn't change	Wrong port selected, or the board isn't printing. Try the port in Device Manager, or test in simulated mode.
Windows protected your PC	**More info**, then **Run anyway** (section 3).
pip: *DLL load failed while importing QtCore*	Your Python is an old Anaconda. Install Python from python.org, or use the .exe.
pip error about a *long path*	Install from a shorter folder, or tick *Disable path length limit* at the end of the python.org installer.

Getting help

Help in the program repeats the main steps. If the program shows an error, details are saved to `%APPDATA%\Wind Tunnel Logger\error.log` (type that into the File Explorer address bar). Send that file, and a screenshot of the window, to your TA.